

Møller Polarimeter Status

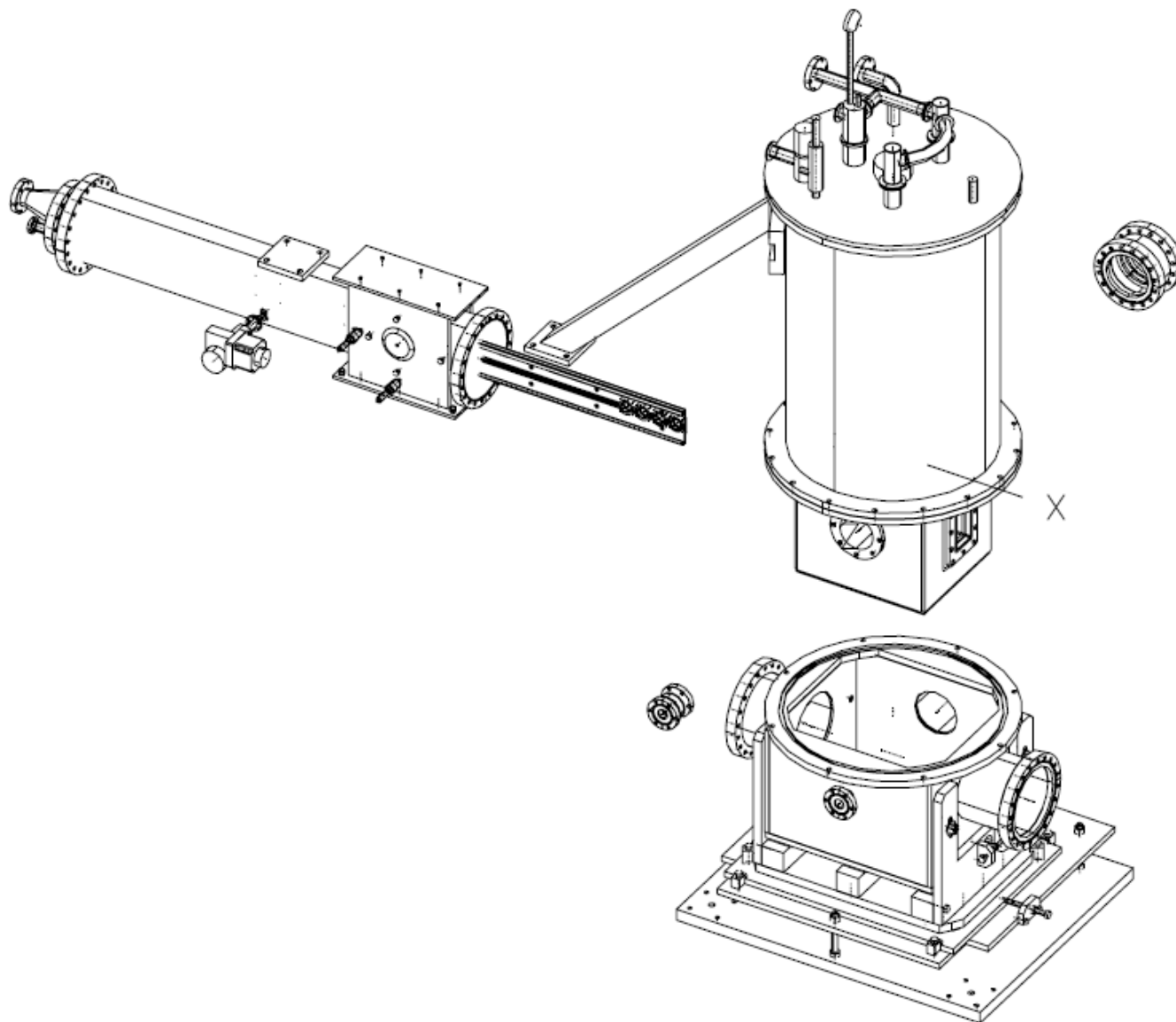
O. Glamazdin ¹

¹ *NSC Kharkov Institute of Physics and Technology,
Kharkov, Ukraine*

What has to be done?

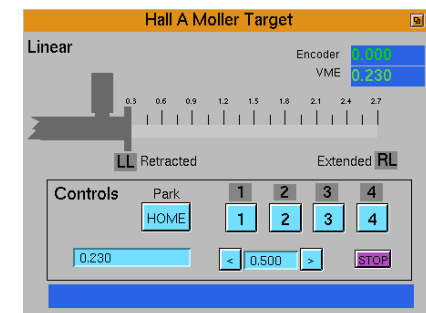
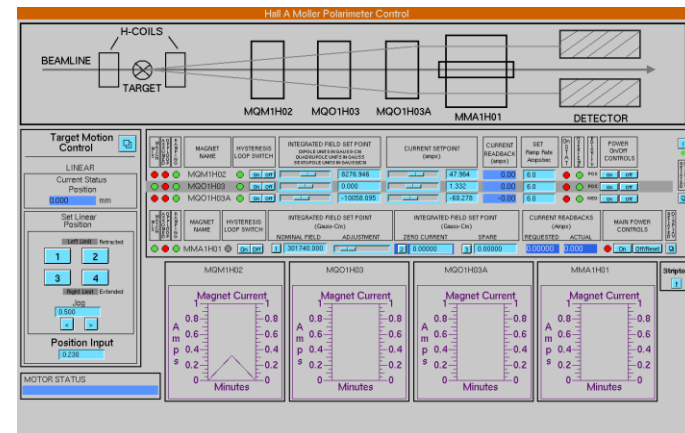
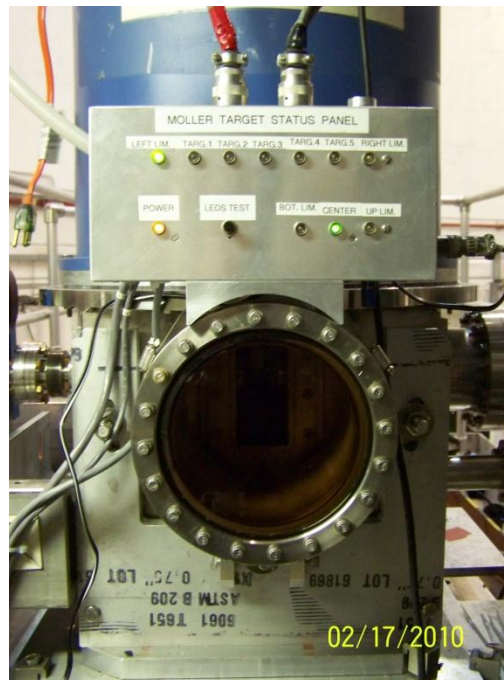
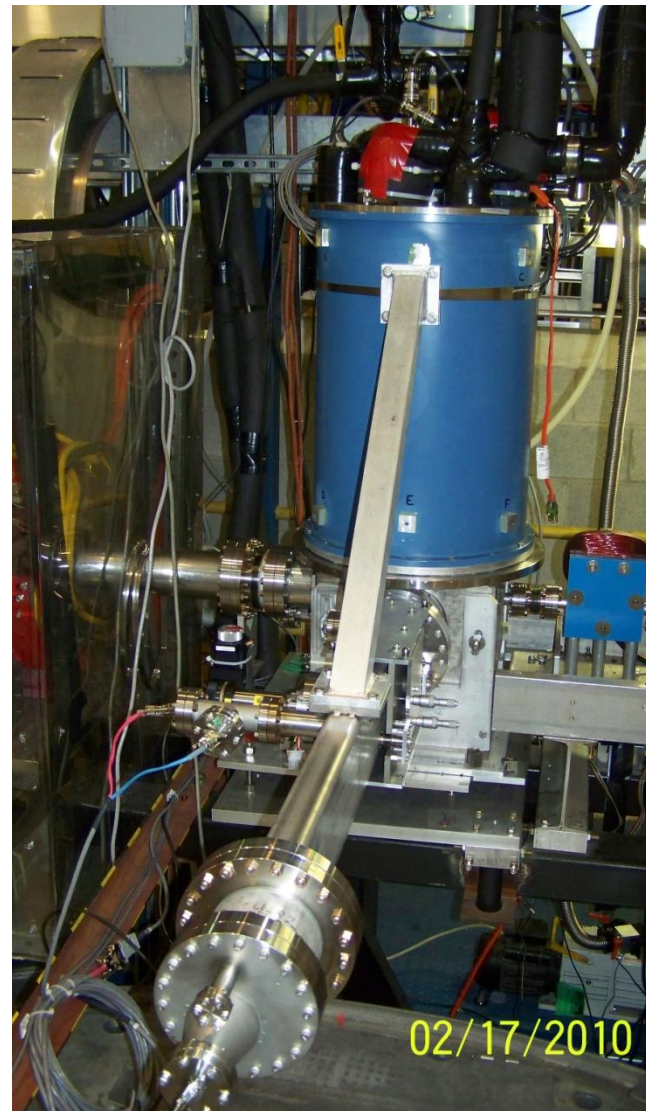
- 1. Polarized electron target "brute force" 3T (100% target magnetization?)*
- 2. Segmented aperture detector (higher rate)*
- 3. Introduce a beam duty cycle (reduce heating)*
- 4. New fast DAQ based on FADC (higher rate, smaller dead time, more information etc.):
Brad, Mindy, Ahmed*

New Møller Target Design



Møller Target Area

MCC motor control code was upgraded
Motor movement was tested
Encoder was tested
FSD was tested
Have to repeat the test with 3T filed



Superconducting Magnet



After reconstruction:

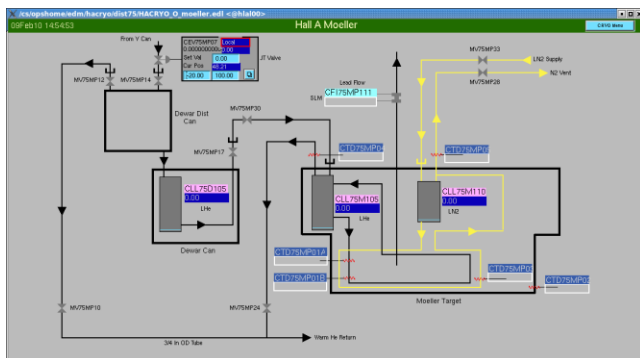
- *Better alignment*
- *Larger N2 outlet diameter*
- *Better IR shield*
- *More N2 and He temperature and level sensors*
- *Vacuum and cooling tests were done*
- *Field mapping was done*
- *Alignment was done*
- *LN2 automatic refill*
- *Cryo system control is ready*
- *Magnet power control is ready*
- *Magnet power cable is not connected*
- *LHe line is not ready*

Waiting for cryo test in the Hall

Superconducting Magnet

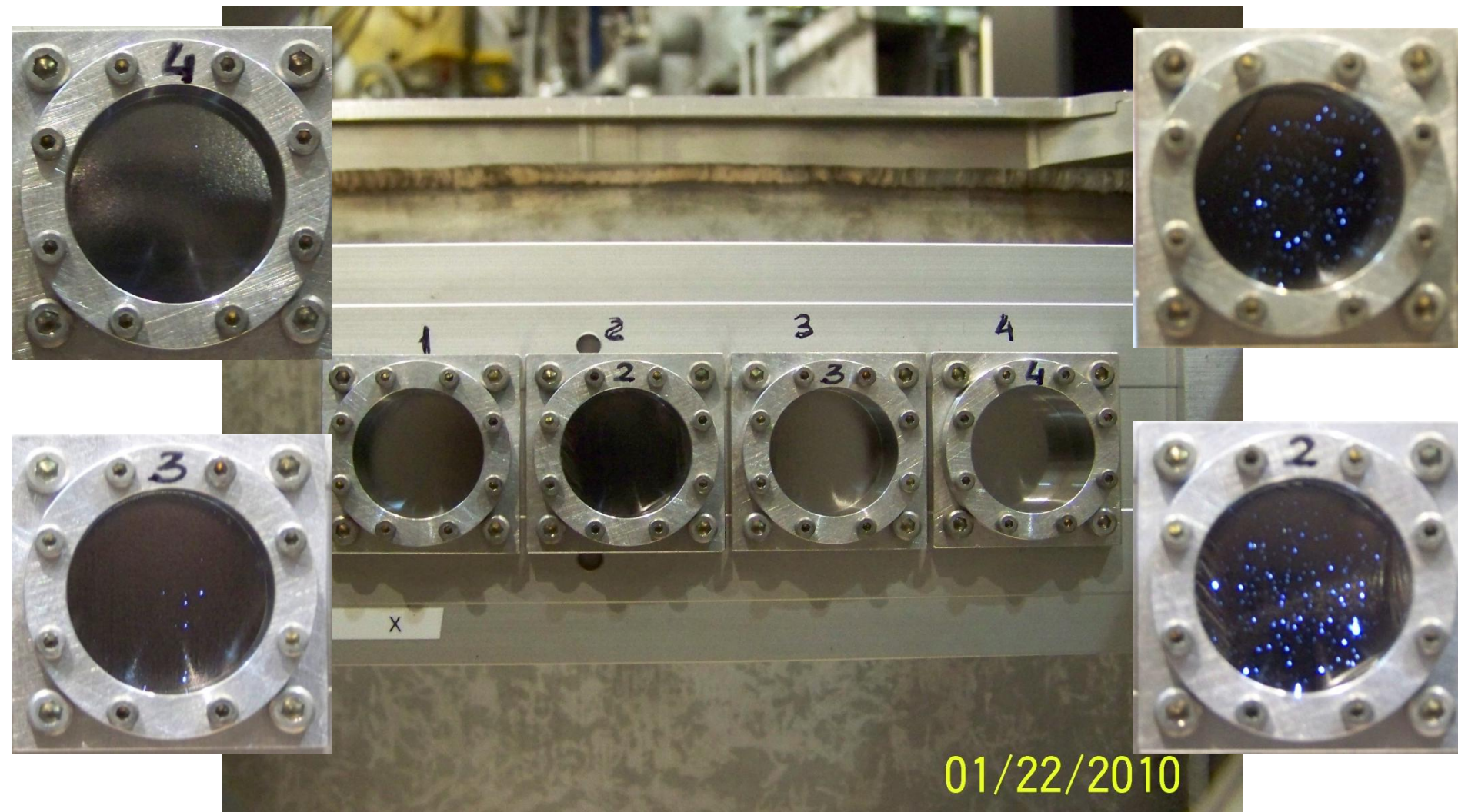


- *LHe manual refill:
~5 days with magnet OFF
~2 days with magnet at 4T*
- *check and refill LHe with any controlled access to the Hall*
- *45min for magnet ramp up*
- *45min for magnet ramp down*
- *software polarity switch (but 1.5hours to change polarity)*



*~1hours for Møller settings +
~1hours to restore the beam
parameters*

Target Foils



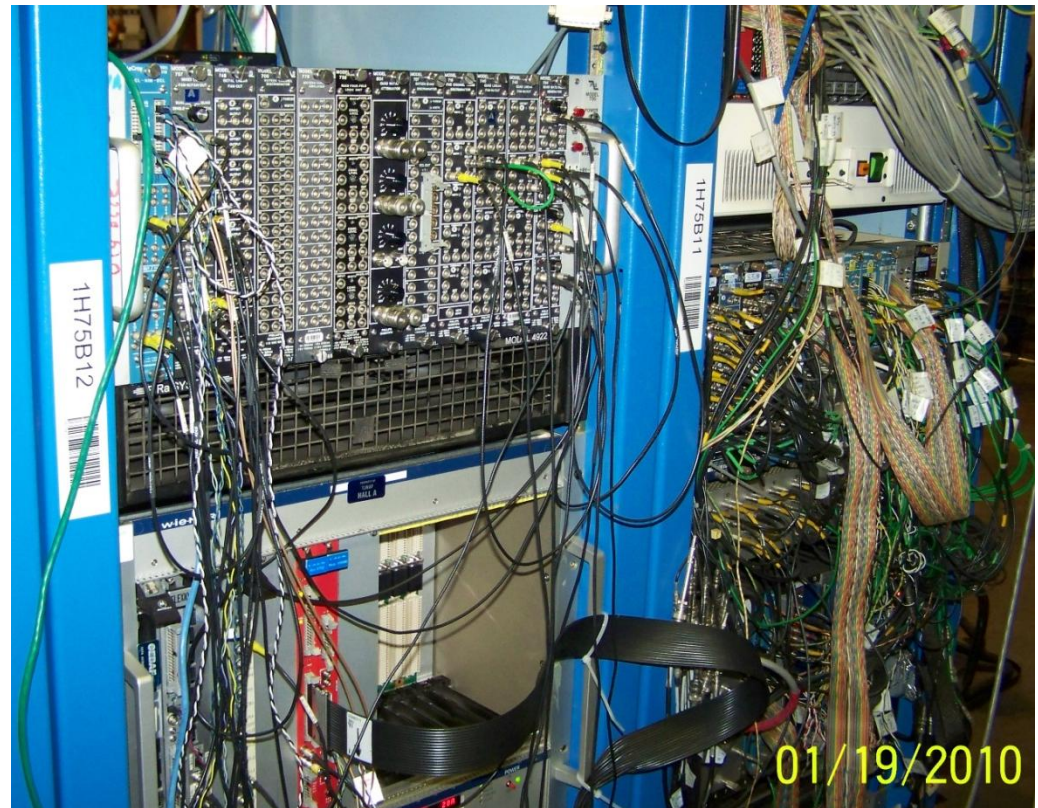
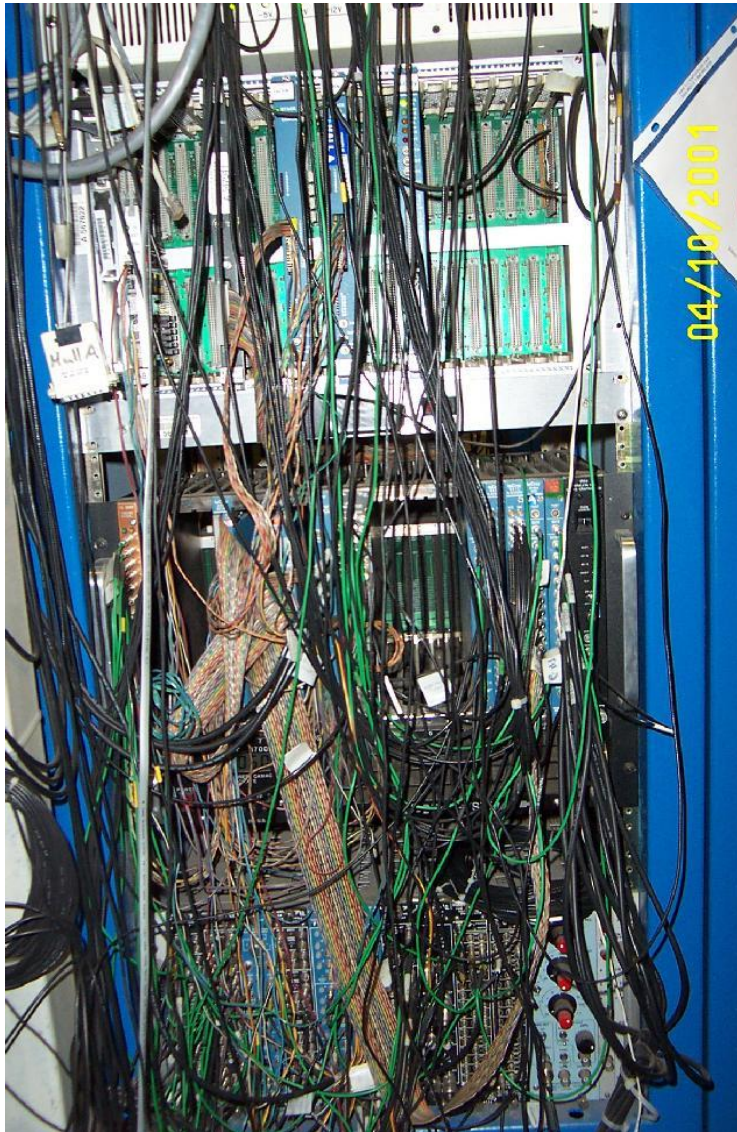
Targets: pure Fe foils 1-1 μ m, 2-1 μ m, 3-4 μ m, 4-12 μ m

Segmented Aperture Detectors



Will be installed next week

Old and New Møller DAQ



For PREX old and new DAQs will be working in parallel

Measurement at High Current Beam

Measurement at high beam current $0.5\mu\text{A} \rightarrow 50\mu\text{A}$?

Reasons:

- no laser phase dependence (fully open slit) $\geq 5\mu\text{A}$;
- working beam position fast-feedback $\geq 5\mu\text{A}$;
- working energy fast-feedback $\geq 5\mu\text{A}$;
- no bleed through effect (time by time check)
 $\geq 5\mu\text{A}$ is "high current beam"

Problems:

- target heating;
- detector counting rate

Measurement with *CW* Beam

25K heating $\rightarrow$ 0.3% target depolarization

Heating by $1.5\mu\text{A}$ CW:

$\Delta T \sim 22\text{K}$ (no raster)

$\Delta T \sim 12\text{K}$ (raster $1\times 1\text{mm}^2$)

Measurement with *CW* Beam

25K heating $\rightarrow$ 0.3% target depolarization

Heating by $1.5\mu\text{A}$ CW:

$\Delta T \sim 22\text{K}$ (no raster)

$\Delta T \sim 12\text{K}$ (raster $1 \times 1\text{mm}^2$)

For PREX:

Raster $1 \times 0.5\text{mm}^2$ and $3\mu\text{A}$ CW ($< 5\mu\text{A}$):

$\Delta T \sim 30\text{K}$

$1\mu\text{m} / 3\mu\text{A} \sim 1\text{hour}$ (0.3%stat error)

$4\mu\text{m} / 3\mu\text{A} \sim 20\text{min}$ (0.3%stat error)

$12\mu\text{m} / 1\mu\text{A} \sim 20\text{min}$ (0.3%stat error)

$\sim 2\text{hours}$

Measurement with *User Mode*

"User mode" - a few ms pulse for each 30ms helicity window

For PREX:

Raster $1 \times 0.5 \text{ mm}^2$ +

10ms of 30ms helicity pulse (30Hz)

(or $\sim 3 \text{ ms}$ of $\sim 8 \text{ ms}$ helicity pulse 120Hz)

$1 \mu\text{m}$ / $10 \mu\text{A}$ beam: $\sim 45 \text{ min}$ for 0.3% stat.er.

Measurement with *Beat Frequency Mode*

Micro-suppression (the beat frequency)

M. Poelker Phys.Rev.ST Accel.Beams 10:053502,2007 or J Musson JLAB-ACE-07-693, Jun 2007. 3pp.

Pre-buncher must be off → **INVASIVE!!!**

Fast energy feed-back is not available!

by Eugene: $n=4$ ($500\text{MHz}/4=125\text{MHz}$)

For PREX: $n \geq 10$ ($500\text{MHz}/10=50\text{MHz}$)

Measurement with *Beat Frequency Mode*

Micro-suppression (the beat frequency)

M. Poelker Phys.Rev.ST Accel.Beams 10:053502,2007 or J Musson JLAB-ACE-07-693, Jun 2007. 3pp.

Pre-buncher must be off → INVASIVE!!!

Fast energy feed-back is not available!

by Eugene: $n=4$ ($500\text{MHz}/4=125\text{MHz}$)

For PREX: $n \geq 10$ ($500\text{MHz}/10=50\text{MHz}$)

For test only:

Raster $1 \times 0.5\text{mm}^2$ +

$n=10$ (micro-suppression) +

10ms of 30ms helicity pulse ("user mode"):

- $1\mu\text{m}$ target/ $50\mu\text{A}$ beam: $\sim 1.5\text{hours}$ for 0.3% stat.er.

- $4\mu\text{m}$ target/ $25\mu\text{A}$ beam: $\sim 45\text{min}$ for 0.3% stat.er.

- $12\mu\text{m}$ target/ $10\mu\text{A}$ beam: $\sim 35\text{min}$ for 0.3% stat.er.

$\sim 3.5\text{hours}$

Møller Run for PREX

Møller settings (+magnet ramp up): *~1hour*

Raster $1 \times 0.5 \text{mm}^2$

CW: (low current $< 5 \mu\text{A}$)

$1 \mu\text{m} / 3 \mu\text{A} \sim 1 \text{hour}$ (0.3%stat error)

$4 \mu\text{m} / 3 \mu\text{A} \sim 20 \text{min}$ (0.3%stat error)

$12 \mu\text{m} / 1 \mu\text{A} \sim 20 \text{min}$ (0.3%stat error)

Target thickness and condition systematic

CW: *~2hours*

"User mode": (high current $> 5 \mu\text{A}$)

10ms of 30ms helicity pulse (30Hz)

$1 \mu\text{m} / 10 \mu\text{A}$ beam: $\sim 45 \text{min}$ for 0.3% stat.er.

"User mode": *~1hour*

Restore PREX beam settings: *~1hour*

Total: *~5hours*

Systematic Error

Variable	Hall C	Hall A	
		<i>Present</i>	<i>Upgrade</i>
<i>Target polarization</i>	0.25%	1.8%	0.5%
<i>Target angle</i>	0.0%	0.5%	0.0%
<i>Analyzing power</i>	0.24%	0.3%	0.3%
<i>Levchuk effect</i>	0.3%	0.2%	0.2%
<i>Target temperature</i>	0.05%	0.0%	0.02%
<i>Dead time</i>	?	0.3%	0.3% !
<i>Background</i>	?	0.3%	0.3% !
<i>Others</i>	0.1%	0.3%	0.5% !
<i>Total</i>	0.47%	2.0%	~1.0%